

Credit Risk EDA Dashboard

Overview

The **Credit Risk EDA Dashboard** is an interactive dashboard developed using **R Shiny** for Exploratory Data Analysis (EDA) of a credit risk dataset. The application provides interactive visualisations, summary statistics, and data quality analysis to support the development of credit risk models.

This project was developed as part of a Credit Risk Modelling project and is currently a **work in progress**. Additional modelling and evaluation features will be added in future updates.

Features

- Interactive R Shiny Dashboard
 - Home Dashboard
 - Data Quality Assessment
 - Univariate Analysis
 - Bivariate Analysis
 - Interactive Visualisations
 - Summary Statistics
 - Correlation Analysis
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Technologies Used

- R
 - R Shiny
 - shinydashboard
 - ggplot2
 - plotly
 - dplyr
 - DT
 - DataExplorer
 - corrplot
-

Project Structure

```
CreditRiskEDA/  
|  
├─ app.R  
└─ Global.R
```

```
├── CreditRiskEDA.Rproj
├── loan_book.csv
├── README.md
├──
├── R/
│   ├── home.R
│   ├── quality.R
│   ├── univariate.R
│   ├── bivariate.R
│   └── insights.R
└── .gitignore
```

Requirements

Before running the project, ensure you have the following installed:

- R (version 4.0 or later)
- RStudio (recommended)

Required Packages

Run the following command in the R Console to install the required packages:

```
install.packages(c(
  "shiny",
  "shinydashboard",
  "ggplot2",
  "plotly",
  "dplyr",
  "DT",
  "DataExplorer",
  "corrplot",
  "tidyr",
  "stringr",
  "readr",
  "scales",
  "scorecard",
  "caret",
  "ROCR",
  "pROC",
```

```
"reshape2"  
)
```

How to Run the Project

1. Download or clone this repository.
2. Open the project folder in **RStudio**.
3. Install the required packages if they are not already installed.
4. Open each of the following files **one at a time**:
5. `Global.R`
6. `R/home.R`
7. `R/quality.R`
8. `R/univariate.R`
9. `R/bivariate.R`
10. `R/insights.R`
11. For **each file**, press **Ctrl + A** to select all the code and then click **Run** (or press **Ctrl + Enter**) to execute the entire file.
12. After all the above files have been executed successfully, open `app.R` and click **Run App** (or run `shiny::runApp()` in the R Console).
13. The Credit Risk EDA Dashboard should launch in your default web browser.

Current Status

Work in Progress

The dashboard currently focuses on Exploratory Data Analysis (EDA). Future updates will include:

- Weight of Evidence (WoE)
 - Information Value (IV)
 - Logistic Regression
 - Feature Engineering
 - ROC Curve
 - AUC Analysis
 - Confusion Matrix
 - Model Performance Evaluation
 - Additional Interactive Dashboard Features
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License

This project is intended for educational and portfolio purposes.